MIL-R-37221
30 August 1976

MILITARY SPECIFICATION

RETINOSCOPE, COPELAND, CORD TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a streak type retinoscope of the Copeland design, suitable for use in determining refractive errors of the eye.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Fiberboard.
PPP-B-676	- Boxes, Set-Up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. All materials employed in the fabrication of the retinoscope shall be the best quality suited for the intended purpose.

3.1.1 Glass parts. All glass parts shall be of the highest quality. The glass shall be free from any defects which may interfere with good optical imagery. Glass surfaces shall be ground and polished to the necessary accuracy for a high standard of performance.

3.1.2 Other materials. Other materials used in the fabrication of the retinoscope shall be of high quality, and shall be suitable in all respects for the purpose intended.

3.2 Description. The retinoscope shall consist of a head assembly, condensing lens assembly, handle assembly, illuminating system, conducting cord, and carrying case.

3.2.1 Head assembly. Retinoscope head shall be attached to the condensing lens assembly by means of screws through the backplate of the head assembly, engaging in threaded holes in the assembly head female bracket (bracket being friction fitted to

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the upper portion of the condensing lens assembly. A chromium plated circular brow-rest and an attached backplate, having a central circular aperture of suitable size, shall enclose the proximal side of the head. A metal hood, essentially hemispheric in shape, shall enclose the distal side of the head assembly. The hood shall be firmly attached to, and removable from, the rim of the brow-rest. Internal and external surfaces of the hood and backplate shall be finished in a dull black color. A mirror bracket attached to the backplate, shall support th e retinoscope mirror essentially centering it with the centerline of the front and back apertures of the head assembly. The mirror shall be circular in shape, and shall be of the front surface type. The reflecting surface shall be inconel applied by evaporation and shall embrace the entire area of the front surface of the mirror, except for a central area, which shall be clear and shall be elliptical in shape. The clear area shall be essentially centered with the centerline of the front and back apertures of the head assembly. The mirror shall be so located in the mirror bracket that the clear central area will appear circular when viewed along the centerline of the front and back apertures of the head. The mirror shall be made of glass, as specified in 3.1.1. With the exception of the optical parts, the interior of the head assembly shall be finished in a dull black color.

3.2.2 Condensing lens assembly. The condensing lens assembly shall consist of a chromium plated metal tube supporting in its upper end a condensing lens. The condensing lens shall be held securely in the mount. The upper end of the tube shall be tapered to fit into the tapered female bracket of the head assembly. The lower portion of the tube shall have a male thread to engage in the thread of the handle assembly. The condensing lens tube shall be made with an elevated knurled area at the lower end, directly above the male thread, and extending around the tube. This knurled area shall provide a suitable finger grip for unscrewing the condensing lens assembly from the handle assembly for replacing the lamp.

3.2.3 Handle assembly. The handle shall consist of a metal tube which shall act as the lamp housing and shall contain the internal electrical assembly. The upper end of the handle shall be chromium plated and shall contain a female thread to fit over the aforementioned male thread of the condensing lens assembly. The lower portion of the handle shall be covered with imitation leather, plastic, or other material suitable to provide a firm hand grip. As an alternate, the lower portion of the handle shall have a knurled area suitable to provide a firm hand grip.

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The internal electrical assembly shall consist of a bakelite or other suitable insulating material cylinder which fits loosely into the handle and which contains the lamp base receptacle to accommodate the lamp hereinafter specified, a bakelite or other suitable insulating material cylinder containing a male plug for connection to the cord set hereinafter described, the two cylinders being located at the upper and lower portions of the handle, respectively, their electrical components being connected by two insulated flexible braided copper wires. A chromium plated metal sleeve shall be fitted over the upper portion of the handle. The sleeve shall be fastened to the insulated portion of the lamp receptacle. The sleeve shall be made with an elevated knurled area at its center, extending around the sleeve, this knurled area providing a suitable thumb and forefinger grip for rotating the sleeve and for sliding it longitudinally over the upper portion of the handle. The wall of the sleeve shall be sufficiently thick to provide a support on the thumbnail or fingernail for maintaining the vertical position of the sleeve while rotating the streak of light. The inside of the sleeve shall be partially lined with velvet ribbon or other suitable material to permit a smooth sliding action of the sleeve over the mating portion of the handle. The rotary movement of the lamp receptacle shall be not less than 180°.

3.2.4 Lamp. Lamp shall have a screw base and be of the straight filament type. It shall have a thread to fit the socket located in the handle assembly. The lamp globe shall be free from striations or irregularities that would affect the clarity of the projected streak image. Instruments shall be supplied with two replacement lamps. (see 6.5)

3.2.5 Cord set. The exterior electrical system shall consist of a cord set intended for connecting the retinoscope to a 2 1/2-3 volt power supply. To one end of the cord shall be affixed a female plug, incorporating an "O" ring, to provide a suitable and secure connection between the cord and the male plug inside the instrument handle. When connected, the plug shall extend beyond the end of the instrument handle, and shall provide a suitable finger grip for ready insertion or removal of the plug. The other end of the cord shall be separated into "Y" shaped terminals, to each of which shall be attached a standard telephone type tip jack intended for insertion into a 2 1/2-3 volt plug-in transformer on into the 2 1/2-3 volt outlet of a standard refractive unit. The cord shall be of the insulated twin conductor type and shall be 5 feet (plus or minus 3 inches) long overall.

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3.2.6 Carrying case. The carrying case shall be molded of a durable, resilient rubber-base material or plastic, washable with soap and water. There shall be a form-fitting molded insert in the bottom of the case to accommodate all the components specified above and prevent their shifting and damage during carrying of the complete set and when the set is not in use. The case shall be provided with a piano-type hinge or two suitable hinges located one near each end of the case, and shall have a nylon zipper closure extending along the side opposite the hinge and essentially around both ends.

3.2.7 Rotary movement. The rotary movement of the receptacle shall be not less than 180° when tested according to 4.5.1.

3.2.8 Streak image. With sleeve in lowest position, the streak shall be in focus at not more than 15 inches in front of the instrument head assembly. With sleeve in the highest position, the width of the blur (out of focus streak image) shall be at least 24 mm when projected at a distance of 30 inches in front of the instrument head assembly. The streak when in focus at 30 inches should be essentially straight and not exceed 4 mm in width. Rotation of the sleeve shall cause the streak image to rotate through at least 180°.

3.3 Identification marking. A metal plate permanently die-stamped or etched with the manufacturer's name or registered trade-mark, model number of the retinoscope, and stock number shall be securely affixed to the retinoscope.

3.4 Service data. The contractor shall enclose, with each unit, two copies each of one or more service data booklets. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. The service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. Each part of the unit other than common fasteners, such as nuts and bolts shall be identified in the service data and parts list by the manufacturer's part number.

3.4.1 Approval. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall

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be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

3.4.2 Waiver of review and distribution of manual. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.5 Workmanship. Workmanship shall be first class throughout. Retinoscopes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection as used in this specification, is defined as both examinations (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105.

4.2.2 Examination. The retinoscope shall be examined to determine compliance with all requirements contained in this specification. The inspection level and acceptable quality level (AQL) shall be as indicated in table I. The unit of product for examination is one retinoscope.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective
Major	S-1	1.0
Minor	S-1	1.0

4.2.3 Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test is one retinoscope.

TABLE II. Sampling criteria for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Rotary movement	3.2.7	4.5.1	S-1	1/
Streak image	3.2.8	4.5.2	S-1	1/

1/ Acceptance number shall be zero for all samples sizes.

4.3 Examination. The retinoscope shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables II and III. Examination, however, shall not be restricted to defects listed in table III. Any retinoscope in the sample containing one or more defects shall be rejected and if the number of defective retinoscopes in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Electrical connections do not mate readily or do not maintain proper contact as revealed by flickering light.
102	Carrying case does not properly accommodate the retinoscope and cord set.
103	Sleeve does not exhibit smooth sliding action over the rotating portion of the handle.
Minor	
201	Protective and decorative finishes not uniformly and meticulously applied.
202	Identification marking of retinscope missing or illegible.

4.4 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packaging and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Rotary movement. The lower end of the instrument handle shall be held in the right hand. The thumb of the right hand is flexed and the thumbnail is placed under the lower edge of the sleeve. The sleeve shall be rotated with thumb and forefinger of the left hand.

4.5.2 Streak image. The ability of the retinoscope to project a streak shall be tested with the sleeve in the lowest position and project a clean streak at not more than 15 inches from the front of the instrument head assembly. With the sleeve in the highest position, the width of the blur (out of focus streak image) shall be at least 24 mm when projected at a distance of 30 inches from the front of the instrument head assembly.

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The streak when in focus at the 30-inch distance shall not exceed 4 mm in width. While the streak image focus is maintained, the sleeve shall be rotated causing the image to rotate through at least 180°. Flare at the outer ends of the filament image or a bright circle surrounding the image shall not be cause for rejection.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative. Each retinoscope shall be thoroughly cleaned and dried in accordance with applicable method as outlined in MIL-P-116. Preservative shall not be applied.

5.1.1.2 Unit package. Each retinoscope in carrying case shall be further packaged in accordance with MIL-P-116, method IC-1.

5.1.2 Level C. Each retinoscope shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Forty-eight (48) retinoscopes shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, water-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Forty-eight (48) retinoscopes shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, overseas type; PPP-B-621, class 1 or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. Retinoscopes shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package and exterior container. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The retinoscope is used to determine the objective refractive quality of the eye.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National Stock Number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-299-8338	RETINOSCOPE, Copeland, Cord Type

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 Replacement lamps. For replacement lamp, order NSN 6240-00-299-8635. Obtain in accordance with appropriate Military Service Management Data Lists.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1937

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-37221 RETINOSCOPE, COPELAND, CORD TYPE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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